Math is Ingenuity

Use ratios and models to explain a clever fix for a real problem.

Name _____ **Date** _____

Tonight's spark

A bike shop ticket lists two gears. The pedal gear has 48 teeth. The back wheel gear has 32 teeth. You pedal around one time. How many times does the back wheel spin?

Make a guess first: once, twice, or somewhere in between?

Try it

Finish each sentence in your own words.

1. **My clever idea solves the problem of ___ by ___.**

   My words _____

2. **One rotation of the pedals gives the back wheel ___ rotations.**

   My words _____

3. **The ___-tooth gear turns when the ___-tooth gear turns.**

   My words _____

4. **The ratio of pedal teeth to wheel teeth is ___:___.**

   My words _____

5. **48:32 and ___:___ are equivalent because I divided both by ___.**

   My words _____

Talk about it

Who can have a clever idea about a problem?

Home mission

Find two jar lids, one big and one small. Stand them on the table with their rims touching. Put a sticky note on the top of each. Turn the big lid around once. Ask your student to predict first: will the small lid turn more than once?

If we get stuck, ask

- How many teeth does each gear have?
- What happens with one full turn of the pedals?
- Can you write that comparison as a ratio?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/10-4/homework.html

How did it feel? (optional)

Tricky Getting there Got it